

QWEN

QWEN.md — Qwen Code context for this module

This file is read by Qwen Code as its module-context file. It is the Qwen Code counterpart of CLAUDE.md and AGENTS.md for this module, and it is a pointer: there is one canonical agent-instruction file per scope.

Read CLAUDE.md — it is mandatory

This module's canonical agent-instruction file is CLAUDE.md in this directory. Before doing any work in this module, open and read CLAUDE.md and this module's CONSTITUTION.md in full. Every rule there binds Qwen Code exactly as it binds Claude Code.

This file is a plain-text pointer and deliberately uses no auto-import directive. Qwen Code's memory-import processor resolves import-prefixed tokens recursively, and the instruction files reference tokens that are not files. To stay compatible with Qwen Code this file contains no such tokens — read CLAUDE.md directly.

INHERITED FROM constitution/ CLAUDE.md

This module's CLAUDE.md inherits, unconditionally, every rule in constitution/CLAUDE.md and the constitution/Constitution.md it references — the HelixConstitution submodule mounted at the parent project's constitution/ directory (resolve the path with constitution/find_constitution.sh from the parent project root). Qwen Code **MUST NOT** weaken any inherited rule.

Anti-Bluff — read first

Tests and Challenges exist for exactly one purpose: to confirm a feature genuinely works for a real end user, end-to-end. A test that passes while the feature is broken is a bluff test and is forbidden. CI green is necessary, never sufficient. See this module's CLAUDE.md, AGENTS.md, and CONSTITUTION.md for the full Sixth/Seventh Law and section 6.J / 6.L mandate.

§11.4.78 — CodeGraph code-intelligence mandate

Inherited by §11.4.78 ID reference from constitution/Constitution.md §11.4.78 (this module's CLAUDE.md and CONSTITUTION.md carry the full anchor with the package name and install commands). In brief: every project worked on by AI coding agents MUST install, initialize, and use CodeGraph — a local semantic code-knowledge-graph exposed to agents over MCP — wired into every CLI agent the developers use, covered by an anti-bluff verification suite. See CLAUDE.md and CONSTITUTION.md in this module, and the constitution submodule Constitution.md §11.4.78, for the full mandate.

§11.4.69 — Universal Sink-Side Positive-Evidence Taxonomy + Mechanical Enforcement (cascaded from constitution submodule §11.4.69)

Verbatim user mandate (2026-05-20): *"THIS MUST HAPPEN NEVER AGAIN!!! We MUST HAVE this all working! Not just for audio but for every single piece of the System!!! Proper full automation when executed with success MUST MEAN that manual testing will be as much positive at least regarding the success results! ... Solution MUST BE universal, generic that solves working flows for all System components and for all future and all existing projects! ... Everything we do MUST BE validated and verified with rock-solid proofs and anti-bluff policy enforcement and fulfillment!"*

Universal generalisation of §11.4.68 (audio-specific) across every user-visible feature class. Every user-visible feature MUST map to one entry in the closed-set §11.4.69 sink-side evidence taxonomy (audio_output, audio_input, video_display, network_throughput, network_connectivity, bluetooth_a2dp, bluetooth_pair, touch_input, sensor, gpu_render, storage_read, storage_write, mediacodec_decode, mediacodec_encode, miracast, cast, boot_service, package_install,

permission_grant, wifi_link, wifi_throughput, ethernet_link, display_topology, drm_playback, subtitle_render — open to additions, never contraction). Every PASS for a feature in the taxonomy MUST cite a captured-evidence artefact path matching the required evidence shape. New helper contracts (additive during grace, mandatory after 2026-06-19): `ab_pass_with_evidence` <description> <evidence_path> (verifies path exists + non-empty), `ab_skip_with_reason` <description> <closed-set-reason> (reasons: `geo_restricted`, `operator_attended`, `hardware_not_present`, `topology_unsupported`, `network_unreachable_external`, `feature_disabled_by_config`; forbids `network_unreachable_external` for any taxonomy feature with a sink-side probe); bare `ab_pass` deprecated (WARN pre-grace, FAIL post-grace). Three pre-build gates + paired §1.1 mutations: CM-SINK-EVIDENCE-PER-FEATURE, CM-NO-FAIL-OPEN-SKIP, CM-AB-PASS-WITH-EVIDENCE-EVERYWHERE. No escape hatch — no `--skip-evidence`, `--config-only-pass`, `--allow-fail-open-skip`, `--legacy-ab-pass-permitted` flag.

Cascade requirement: This anchor (verbatim or by §11.4.69 reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Propagation gate CM-COVENANT-114-69-PROPAGATION enforces the anchor literal across the consumer fleet; paired mutation strips the literal → gate FAILs. Severity-equivalent to a §11.4 PASS-bluff at the sink-side-evidence layer. **Canonical authority:** constitution submodule Constitution.md §11.4.69 for the full mandate.

§11.4.75 — Mechanical Enforcement Without Exception (cascaded from constitution submodule §11.4.75)

Verbatim user mandate (2026-05-20): *"Why do these violations still happen!? This is a serious problem! We cannot rely on stability nor consistency if we cannot respect our Constitution, mandatory rules and constraints! Is there a way to make this always respected, followed and applied without exception fully and unconditionally!? WE MUST HAVE THIS WORKING FLAWLESSLY!!! Do investigate the root causes of such problems! Once all problems are identified WE MUST apply proper mechanisms for this not to happen NEVER EVER AGAIN!"*

The §11.4 covenant historically relied on agent + operator vigilance; three 2026-05-19→20 forensic incidents proved that late-binding enforcement fires hours-to-days after the violator commit reaches every remote. §11.4.75 closes the gap with FIVE independent mechanical enforcement layers — bypassing any single layer does not bypass the discipline: (1) local pre-commit git hook (refuses staged .md lacking sibling .html+.pdf); (2) `commit_all.sh` integration (`_constitution_sibling_check` +

auto-sync_all_markdown_exports.sh self-repair); (3) local pre-push git hook (re-runs siblings + propagation-gate subset); (4) post-commit auto-repair hook (auto-generates orphan-.md siblings, idempotent + recursion-guarded); (5) local-only final-gate ritual (remote CI DISABLED per User mandate — operator runs pre_build_verification.sh + meta-test before every tag per §11.4.40). Helper contracts: scripts/install_git_hooks.sh, scripts/git_hooks/{pre-commit,pre-push,post-commit,commit-msg}, _constitution_sibling_check. The commit-msg hook enforces a Bypass-rationale: <reason> footer when --no-verify is detected; docs/audit/bypass_events.md accumulates the audit trail. Five gates with paired §1.1 mutations: CM-COVENANT-114-75-PROPAGATION, CM-GIT-HOOKS-INSTALL-SCRIPT, CM-GIT-HOOKS-SOURCE-DIR, CM-COMMIT-ALL-SIBLING-CHECK, CM-CI-WORKFLOW-PRESENT. No escape hatch — no --skip-hooks, --bypass-enforcement, --allow-orphan-md, --ci-not-applicable, --mechanical-enforcement-not-needed flag.

Cascade requirement: This anchor (verbatim or by §11.4.75 reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Propagation gate CM-COVENANT-114-75-PROPAGATION; paired mutation strips the literal → gate FAILs. Severity-equivalent to a §11.4 PASS-bluff at the enforcement layer. **Canonical authority:** constitution submodule Constitution.md §11.4.75 for the full mandate.

§11.4.76 — Containers-Submodule Mandate (cascaded from constitution submodule §11.4.76)

Verbatim user mandate (2026-05-20): *"For any work or requirements of running services or codebase inside the Containers (Docker / Podman / Qemu / Emulators, and so on) we MUST USE / INCORPORATE the Containers Submodule properly: <https://github.com/vasic-digital/containers> (git@github.com:vasic-digital/containers.git). Containers Submodule contains all means for us to Containerize our code and services! If any feature or Containing System is missing or not supported we MUST EXTEND IT properly like we do all of our projects! No bluff work is allowed of any kind!"*

For ANY containerized workload (Docker / Podman / Qemu / Kubernetes / container-backed emulators), every consuming project MUST: (1) install vasic-digital/containers (digital.vasic.containers) as a Git submodule; (2) consume via replace directive during development + pinned commit SHAs in production; (3) boot infra on-demand via pkg/boot + pkg/compose + pkg/health so operators are never required to start podman machine / docker compose up manually — the boot is part of the test entry point (the on-demand-infra invariant); (4) extend the Submodule (PR upstream) for missing runtimes / lifecycle primitives — never

reimplement in-project (per §11.4.74); (5) anti-bluff: integration tests claiming to exercise containerized components MUST actually boot them via the Submodule — short-circuit fakes that bypass boot are a §11.4 violation. Tracker rows touching containerization MUST record Catalogue-Check: extend vasic-digital/containers@<sha> (or reuse). Planned gate CM-CONTAINERS-USED scans container-touching PRs for digital.vasic.containers/... imports; paired mutation strips the import + asserts FAIL.

Cascade requirement: This anchor (verbatim or by §11.4.76 reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Propagation gate CM-COVENANT-114-76-PROPAGATION; paired mutation strips the literal → gate FAILs. **Canonical authority:** constitution submodule Constitution.md §11.4.76 for the full mandate.

§11.4.77 — Regeneration-Mechanism-Required Mandate (cascaded from constitution submodule §11.4.77)

Verbatim user mandate (2026-05-20): *"We must be sure that after excluding anything from Git versioning we still have the mechanism which will out of the box obtain or re-generate missing content!"*

Every .gitignore entry excluding (a) >~100 MiB OR (b) any artefact essential to building / running / testing the project MUST carry a documented + automated mechanism to either re-obtain (download from authoritative source: vendor tarball, SDK installer, npm/pip/cargo/go-mod/container registry, dedicated git submodule, S3/GCS) OR re-generate (run from tracked source via build pipeline, code-gen, asset render, captured-evidence replay, container build). Required artefacts per qualifying entry: (1) .gitignore-meta/<entry-slug>.yaml declaring pattern + mechanism-type + script-path + expected-disk-usage + vendor-url-or-source + integrity hash + requires-network + requires-credentials; (2) a non-interactive entry in scripts/setup.sh post-clone bootstrap; (3) a pre-build gate verifying regenerated content present OR a recent .gitignore-meta/.regenerated/<slug>.ok stamp; (4) README + docs/guides/*.md describing the mechanism + manual fallback + time/disk budget + §11.4.10 credentials. Bare .gitignore additions without the mechanism are a §11.4 PASS-bluff variant — codebase appears complete but a fresh clone cannot build/run. No escape hatch — no --skip-regen-mechanism, --gitignore-is-enough, --operator-already-has-content flag. Planned gate CM-GITIGNORE-REGEN-MECHANISM + paired §1.1 mutation (strip a required YAML key → gate FAILs).

Cascade requirement: This anchor (verbatim or by §11.4.77 reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Propagation gate CM-COVENANT-114-77-PROPAGATION; paired mutation strips the literal → gate FAILs. Severity-equivalent to a §11.4 PASS-bluff at the repository-hygiene layer. **Canonical authority:** constitution submodule Constitution.md §11.4.77 for the full mandate.

§11.4.79 — Own-Org Submodules MUST Be Included in the CodeGraph Index (cascaded from constitution submodule §11.4.79)

Verbatim user mandate (2026-05-21): "All Submodules we use in the project and that are part of organizations to which we have the full access via GitHub, GitLab and other CLIs MUST BE included into the codegraph database and initialized / scanned / synced!"

Refines §11.4.78's exclude-list with a per-submodule-ownership split: (a) own-org submodules (full write access via the project's CLIs — canonical orgs vasic-digital + HelixDevelopment) MUST be INCLUDED in the index; (b) third-party submodules (the §11.4.74 no-match → vendor path) MUST be EXCLUDED. Operational steps: (1) git submodule update --remote --merge to pull latest before re-indexing, respecting load-bearing pins on third-party submodules; (2) adjust .codegraph/config.json exclude list to keep own-org paths in scope; (3) re-index via scripts/codegraph_setup.sh; (4) verify via scripts/codegraph_validate.sh with ≥ 1 probe resolving a symbol living ONLY inside an own-org submodule; (5) paired §1.1 mutation — temporarily add the own-org submodule to exclude → validate MUST FAIL on the cross-submodule probe → restore. An index that lies about reachable symbols is a PASS-bluff against AI agents. Own-org submodules silently excluded without an audit trail in .codegraph/config.json comments is a release blocker.

Cascade requirement: This anchor (verbatim or by §11.4.79 reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Propagation gate CM-COVENANT-114-79-PROPAGATION; paired mutation strips the literal → gate FAILs. **Canonical authority:** constitution submodule Constitution.md §11.4.79 for the full mandate.

§11.4.80 — CodeGraph Regular-Update + Sync Automation Mandate (cascaded from constitution submodule §11.4.80)

Verbatim user mandate (2026-05-21): *"We MUST regularly check for the updates and execute codegraph npm updates so the latest version of it is always installed on the host machine! ... Make sure we have proper full automation bash scripts which will run regularly and that these are part of the constitution Submodule ... Make sure all updates, sync processes we do and important codegraph related events are all documented under docs/codegraph in Status and Status_Summary documents ... and regularly export them like all other Status docs into the PDF and HTML!"*

Three deliverables (all living in the constitution submodule, inherited by reference per §3 — consuming projects invoke at `{CONST_DIR}/scripts/codegraph_*.sh`, never copy): (1) `scripts/codegraph_update.sh` — npm-installs latest `@colbymchenry/codegraph` after a registry version check; appends old/new version to `docs/codegraph/Status.md`; anti-bluff verifies `codegraph --version` reflects the new version after install (npm exit 0 ≠ working binary). (2) `scripts/codegraph_sync.sh` — after a successful update runs `codegraph status` → `codegraph sync` . → `codegraph status` → the project's `scripts/codegraph_validate.sh`; appends every step's output to BOTH the project's and the constitution's `docs/codegraph/Status.md`. (3) `docs/codegraph/Status.md` + `Status_Summary.md` append-only ledgers, exported to `.html` + `.pdf` per §11.4.65. Cadence: weekly floor (per §11.4.45). A consuming project that has not run `codegraph_update.sh` in >2 weeks AND has open AI-agent work is a release blocker. Paired §1.1 mutation: downgrade installed version → script detects drift → restore.

Cascade requirement: This anchor (verbatim or by §11.4.80 reference) MUST appear in every owned submodule's `CONSTITUTION.md`, `CLAUDE.md`, and `AGENTS.md`. Propagation gate `CM-COVENANT-114-80-PROPAGATION`; paired mutation strips the literal → gate FAILs. **Canonical authority:** constitution submodule `Constitution.md` §11.4.80 for the full mandate.

§11.4.81 — Cross-Platform-Parity Mandate (cascaded from constitution submodule §11.4.81)

Verbatim user mandate (2026-05-21): *"Any Linux-only blocker / issue we have MUST BE created macOS and other supported platforms equivalent! So, depending on platform*

*proper implementation will be used for particular OS!
EVERYTHING MUST BE PROPERLY EXTENDED AND UP-
DATED!"*

Every consuming project whose supported-platforms manifest lists more than one OS MUST, for every feature/test/gate/challenge/mutation depending on platform-specific primitives, ship a per-OS-equivalent implementation chosen at runtime via `uname -s` (or equivalent detection). Three sub-mandates: **(A) Per-OS implementation REQUIRED** — Linux `cgroup/systemd//proc` primitives MUST have documented per-OS equivalents (POSIX `setrlimit/ulimit`, macOS `launchd`, BSD `rctl`, Windows Job Object) chosen via runtime dispatch. **(B) Per-OS tests REQUIRED** — every platform-dependent gate test MUST have case `"$(uname -s)"` in branches with positive captured evidence per §11.4.2 + §11.4.5 in each branch; SKIP-with-reason acceptable ONLY when the platform genuinely cannot enforce the invariant. **(C) Honest kernel-gap citation + adjacent equivalent test REQUIRED** — where a Linux primitive has NO equivalent due to a documented kernel limitation (canonical: XNU does not enforce `RLIMIT_AS` for unprivileged processes), the test MUST detect the gap at runtime, SKIP with exact kernel reason + reproducer + honest-gap-doc link, AND provide an ADJACENT test exercising the closest invariant the platform CAN enforce (e.g. `RLIMIT_CPU+SIGXCPU` as the macOS proxy), itself anti-bluff with a paired §1.1 mutation. Gate CM-CROSS-PLATFORM-PARITY scans for case `"$(uname -s)"` blocks asserting a non-SKIP branch (or honest-gap citation) per platform in the manifest; paired mutation strips a Darwin branch → gate FAILs. No escape hatch.

Cascade requirement: This anchor (verbatim or by §11.4.81 reference) MUST appear in every owned submodule's `CONSTITUTION.md`, `CLAUDE.md`, and `AGENTS.md`. Propagation gate CM-COVENANT-114-81-PROPAGATION; paired mutation strips the literal → gate FAILs. Release blocker on multi-platform projects. **Canonical authority:** constitution submodule `Constitution.md` §11.4.81 for the full mandate.

§11.4.82 — Iteration-Speedup Discipline Mandate (cascaded from constitution submodule §11.4.82)

Verbatim user mandate (2026-05-22): "How can we speed-up this whole development and fixing process? ... Do not forget to all speed optimizations critical rules and mandatory constraints MUST BE all added into our root (constitution Submodule) Constitution.md, CLAUDE.md, AGENTS.md and QWEN.md and all other relevant constitution Submodules files!"

Iteration cycle time is a first-order quality enabler. Every consuming project's build / test / commit / debug pipeline **MUST** adopt these speedup disciplines AS MANDATORY (each independently enforceable): (A) Phase-1 forensic (superpowers:systematic-debugging) before any speculative source patch — speculative patches without FACT-grade root cause are §11.4.6 + §11.4.82 violations; (B) Live-ADB-First (or live-equivalent) before any rebuild — strengthens §11.4.51 to a release-blocker mandate; (C) 30-second pre-flight before launching rebuild orchestrators (device/sink reachability, host memory/disk, no stale locks, no orphan processes); (D) persistent build caches outside containers (ccache/sccache/Gradle daemon bind-mounted to host); (E) module-only rebuild for loadable-module-only changes; (F) parallel multi-device testing with separate qa-results/<TS>/<device-tag>/ outputs; (G) subagent scope discipline + worktree isolation (≤30 min budget, single-responsibility, isolation: "worktree" default); (H) lock-file + stale-process hygiene (clean .git/index.lock, disable auto git-gc in concurrent repos); (I) cycle telemetry per §11.4.24 (commit hash, per-phase wall-clock, speedup-flag set, outcome — aggregated weekly). Gate CM-ITERATION-SPEEDUP-DISCIPLINE audits recent cycles for telemetry citing which of (A)-(I) applied; paired §1.1 mutation strips the speedup-flag column → gate FAILs. No escape hatch — no --skip-phase1-forensic, --no-pre-flight, --rebuild-everything-always, --unlimited-subagent-scope, --ignore-locks, --no-telemetry flag.

Cascade requirement: This anchor (verbatim or by §11.4.82 reference) **MUST** appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Propagation gate CM-COVENANT-114-82-PROPAGATION; paired mutation strips the literal → gate FAILs. Release blocker. **Canonical authority:** constitution submodule Constitution.md §11.4.82 for the full mandate.

§11.4.83 — docs/qa/ End-User Evidence Mandate (cascaded from constitution submodule §11.4.83)

*Verbatim user mandate (2026-05-22): "every feature that ships **MUST** carry a recorded e2e communication transcript + any attached materials under docs/qa/<run-id>/ (per-feature subdirectories). A feature with no QA transcript is itself a §107 PASS-bluff — it claims to work but has no auditable runtime evidence. Bot-driven automation **MUST** preserve full bidirectional communication threads as proof."*

Every feature that ships **MUST** carry a recorded end-to-end communication transcript plus any attached materials (screenshots, request/response payloads, audio, file uploads) committed under docs/qa/<run-id>/ — one directory per feature run. Operative rule: (1) every consuming project **MUST** maintain a docs/qa/ tree, each

new feature under docs/qa/<run-id>/ where <run-id> is monotonic + greppable (timestamp / ATM-NNN / other workable-item ID per §11.4.54); (2) transcripts MUST be full bidirectional — every prompt/command sent + every response received (one-sided is not a transcript); (3) attached materials MUST be committed in-repo (no external-only links — that is a §11.4.13 sink-side violation); (4) bot-driven / agent-driven QA automation MUST preserve the full conversation thread as the proof artefact; (5) release gates MUST refuse to tag a version that has any feature-shipping commit without its matching docs/qa/<run-id>/ directory. A feature with no QA transcript is a §11.4 / §107 PASS-bluff. Composes with §11.4.2 / §11.4.5 / §11.4.13 / §11.4.65 / §11.4.69 / §11.1.

Cascade requirement: This anchor (verbatim or by §11.4.83 reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Propagation gate CM-COVENANT-114-83-PROPAGATION; paired mutation strips the literal → gate FAILs. Release blocker — no --qa-evidence-optional escape hatch. **Canonical authority:** constitution submodule Constitution.md §11.4.83 for the full mandate.

§11.4.84 — Working-Tree Quiescence Rule for Subagent Commits (cascaded from constitution submodule §11.4.84)

Verbatim user mandate (2026-05-22): "no subagent commit may proceed while any concurrent mutation gate is in flight in the same checkout. Before git add, the committing agent MUST grep its own working tree for mutation markers (MUTATED for paired, // always pass, return json.Marshal shortcut paths, etc.). Any unexplained file in the staging area triggers ABORT."

No subagent (or main-thread) commit may proceed while any concurrent mutation gate, paired-mutation experiment, or other in-flight mutation is live in the same checkout. Before git add, the committing agent MUST grep its own working tree for mutation markers (MUTATED for paired, // always pass, return json.Marshal shortcut paths, // MUTATION / # MUTATION annotations, _mutated_* filename suffixes, etc.) and explicitly account for every modified file in the staging area; any unexplained file → ABORT. (Forensic case: a logo-fix subagent's git add swept an // always pass JWT-verify mutation residue into an unrelated commit pushed to all four mirrors — a real security-defect window.) Operative rule: (1) pre-git add greps for mutation markers + cross-checks git status --porcelain against the subagent's declared scope; unaccounted entries → ABORT; (2) any active mutation gate MUST be serialised (mutate → assert FAIL → restore → assert PASS) and the working tree verifiably clean before any unrelated commit; (3) concurrent subagents in the SAME checkout MUST coordinate through a

lockfile (.git/MUTATION_IN_PROGRESS) — cleaner solution is git worktree add per subagent (composes with §11.4.20/§11.4.70); (4) post-commit mutation-residue-scanner MUST run before push — any commit containing a mutation marker → push BLOCKED.

Cascade requirement: This anchor (verbatim or by §11.4.84 reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Propagation gate CM-COVENANT-114-84-PROPAGATION; paired mutation strips the literal → gate FAILs. A mutation marker that lands in a tagged commit is a critical defect regardless of how briefly it persisted. **Canonical authority:** constitution submodule Constitution.md §11.4.84 for the full mandate.

§11.4.85 — Stress + Chaos Test Mandate (cascaded from constitution submodule §11.4.85)

Verbatim user mandate (2026-05-24): *"Every fix or improvement you do MUST BE covered with full automation stress and chaos tests so we are sure nothing can break the functionality and all edge cases are monitored and polished and additionally fixed if that is needed! Everything must produce rock solid proofs and follow fully no-bluff policy!"*

Every fix or improvement landed MUST ship with full-automation **stress** AND **chaos** test suites exercising edge cases, sustained load, concurrent contention, and failure-injection. Happy-path coverage alone is a §11.4 / §107 PASS-bluff at the resilience layer. **Stress** (closed-set): sustained load ($N \geq 100$ iterations OR ≥ 30 s wall-clock, p50/p95/p99 latency recorded) + concurrent contention ($N \geq 10$ parallel invocations, no deadlock/leak) + boundary conditions (empty/max/off-by-one, each categorised). **Chaos** (closed-set, per fix-class appropriateness): process-death injection + network-fault injection (drop/delay/reorder) + input-corruption injection + resource-exhaustion injection (disk full, OOM, FD exhaustion — refuse cleanly OR degrade, NEVER crash) + state-corruption injection (mid-flight lock loss, partial-write). Every stress + chaos PASS MUST cite a captured-evidence artefact path per §11.4.5 + §11.4.69. Helper library stress_chaos.sh provides ab_stress_run, ab_stress_concurrent, ab_chaos_kill_pid_during, ab_chaos_drop_network_during, ab_chaos_corrupt_file_during, ab_chaos_oom_pressure_during, ab_chaos_disk_full_during, each composing with ab_pass_with_evidence / ab_skip_with_reason. Cleanup non-negotiable in trap '...' EXIT (cleanup failure = §11.4.14 violation). Four-layer coverage per §11.4.4(b) + paired §1.1 mutation (strip chaos-injection or evidence-capture → gate FAILs). No escape hatch — no --skip-stress, --no-chaos, --happy-path-suffices, --stress-test-later flag.

Cascade requirement: This anchor (verbatim or by §11.4.85 reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Propagation gate CM-COVENANT-114-85-PROPAGATION; paired mutation strips the literal → gate FAILs. Release blocker. **Canonical authority:** constitution submodule Constitution.md §11.4.85 for the full mandate.

§11.4.86 — Roster/Corpus-Backed Status-Doc Auto-Sync Mandate (cascaded from constitution submodule §11.4.86)

Verbatim user mandate (2026-05-25): *"Make sure that assets and players Status docs are ALWAYS regularly updated and in sync like all others Status docs — any time we add or modify the assets content(s) or we change or add new / remove existing pre-installed video and audio player apps! This MUST WORK OUT OF THE BOX!"*

Some Status docs (§11.4.45) are backed by a tracked roster (installed apps/components) or a tracked asset corpus (test/media asset directory) rather than narrative alone. Their freshness MUST NOT depend on operator vigilance — the moment a roster/corpus member changes (app added/removed/renamed; asset added/modified/removed) the Status doc + Status_Summary + HTML + PDF MUST resync out of the box, mechanically. Mechanism (all must hold): (1) drift-proof fingerprint — sha256 of the sorted member list (NOT mtime), persisted in a sidecar beside the Status doc; (2) a sync helper that regenerates the fingerprint + re-exports HTML+PDF via the §11.4.65 exporter, wired so sync is automatic; (3) a pre-build gate that FAILs when the live fingerprint differs from the persisted one (mirrors §11.4.12 CM-ISSUES-SUMMARY-SYNC + §11.4.45 sync_integration_status); (4) a paired §1.1 mutation corrupting the fingerprint and asserting the gate FAILs. Classification: universal — the consuming project supplies the specific docs, roster/corpus sources, helper, and gate name per §11.4.35.

Cascade requirement: This anchor (verbatim or by §11.4.86 reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Propagation gate CM-COVENANT-114-86-PROPAGATION; paired mutation strips the literal → gate FAILs. Release blocker — no --skip-roster-sync, --allow-status-drift, --roster-sync-not-applicable flag. **Canonical authority:** constitution submodule Constitution.md §11.4.86 for the full mandate.

§11.4.87 — Endless-Loop Autonomous Work + Zero-Idle Agent Dispatch + Anti-Bluff Testing Mandate (cascaded from constitution submodule §11.4.87)

Verbatim user mandate (2026-05-26): "*continue in endless loop fully autonomously*" (and any semantically-equivalent phrasing).

When the operator instructs an AI agent to continue in an endless autonomous loop, the agent **MUST** treat it as a **HARD-CONTRACT** covenant: (A) continue working until docs/Issues.md Status-column has zero non-terminal entries AND docs/CONTINUATION.md §3 Active work is empty AND no background subagent is mid-execution AND no external dependency is in-flight; (B) dispatch background subagents for parallelisable work — main + every subagent operate concurrently, "waiting for results" is the **ONLY** acceptable idle reason; (C) every closure lands four-layer test coverage per §11.4.4(b) with captured-evidence (audio/video/network/UI/sysfs physical proofs); (D) the §11.4 anti-bluff covenant family (§11.4.1 / §11.4.2 / §11.4.6 / §11.4.7 / §11.4.27 / §11.4.50 / §11.4.52 / §11.4.68 / §11.4.69 / §11.4.83) is the operative truth-discipline — tests AND HelixQA Challenges bound equally; (E) the loop terminates **ONLY** on all-conditions-met, explicit operator STOP, host-session-safety demand, or scheduled wake on a known-future-actionable signal. No escape hatch — no --idle-OK, --skip-endless-loop, --bluff-permitted-for-this-task, --metadata-only-test-suffices, --no-physical-proof-required flag.

Cascade requirement: This anchor (verbatim or by §11.4.87 reference) **MUST** appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Propagation gate CM-COVENANT-114-87-PROPAGATION; paired mutation strips the literal → gate FAILs. Release blocker. **Canonical authority:** constitution submodule Constitution.md §11.4.87 for the full mandate.

§11.4.88 — Background-Push Mandate: Commit-Lock Release Immediately After Commit, Push Runs Detached (cascaded from constitution submodule §11.4.88)

Forensic anchor (2026-05-26): a single commit_all.sh held its flock ~5 hours because do_push ran synchronously after the commit landed — every subsequent commit blocked on a slow mirror push irrelevant to the local commit's durability. Implementation seam

for §11.4.87(B) zero-idle. The mandate: (A) `.git/.commit_all.lock` MUST be released IMMEDIATELY after `git commit` returns 0 — the commit is durable on local disk regardless of remote push outcome; (B) push runs detached via `nohup ./push_all.sh ... > <log> 2>&1 & + disown` — the orchestrator's exit code reports COMMIT success, NOT push success; (C) `push_all.sh` acquires per-remote flock `.git/.push.<remote>.lock` so concurrent invocations targeting the same remote serialize but different-remote invocations run in parallel; (D) backgrounded push failures land in `qa-results/push_failures/<ts>_<remote>.log` — the next autonomous-loop tick checks per §11.4.87(A) "no external dependency in-flight" gate; (E) synchronous-push escape: explicit `--sync-push` CLI flag preserves legacy behaviour for §11.4.41 force-push merge-first audit paths. Gates CM-COVENANT-114-88-PROPAGATION + CM-BACKGROUND-PUSH-WIRED + paired §1.1 mutations. Synchronous push (without `--sync-push`) = §11.4 PASS-bluff at the execution layer.

Cascade requirement: This anchor (verbatim or by §11.4.88 reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Propagation gate CM-COVENANT-114-88-PROPAGATION; paired mutation strips the literal → gate FAILs. Release blocker — no escape hatch beyond `--sync-push` for force-push events. **Canonical authority:** constitution submodule Constitution.md §11.4.88 for the full mandate.

§11.4.89 — Background Test Execution Mandate (cascaded from constitution submodule §11.4.89)

Verbatim user mandate (2026-05-27): *"Any tests we are executing, especially long test cycles, MUST BE performed in background in parallel with main work stream! This MUST NOT block our capabilities to work on queued workable items. Main work stream can be blocked or sit iddle only if absolutely needed and if it depends hard on results of some background execution."*

Symmetric anchor to §11.4.88 (background push) at the test-execution layer. Mandate: (A) long-running tests (>30 s expected: `pre_build`, `meta_test`, `test_all_fixes`, `recent_work_validate`, HelixQA banks, 4-phase cycles, full-suite retests, audio supervisors, dual-display recorders) MUST run via `nohup ... > <log> 2>&1 & + disown` with the log under a known dir (`qa-results/<test_id>_<ts>.log`); (B) the main stream proceeds to the §11.4.42 priority queue immediately; (C) hard-dependency gating — poll an exit-status file or `pgrep -af <test>` before steps that need the exit code, surfacing as §11.4.66 interactive options if the test is still running; (D) failures land in `<log>` files, the next loop tick checks; (E) foreground execution permitted ONLY for <30 s tests OR explicit operator authorisation; (F) per-script flock serialises same-script invocations,

different-script invocations parallel. Gates CM-COVENANT-114-89-PROPAGATION + CM-BACKGROUND-TEST-EXECUTION-WIRED + paired §1.1 mutations.

Cascade requirement: This anchor (verbatim or by §11.4.89 reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Propagation gate CM-COVENANT-114-89-PROPAGATION; paired mutation strips the literal → gate FAILs. Release blocker — no escape hatch beyond explicit per-invocation operator authorisation. **Canonical authority:** constitution submodule Constitution.md §11.4.89 for the full mandate.

§11.4.90 — Obsolete Status + Per-Item Obsolescence Audit (cascaded from constitution submodule §11.4.90)

Verbatim user mandate (2026-05-27): *"Bug No 6 ... seems obsolete after latest request for new behavior ... mark obsolete tickets with some light gray background ... text - the description to be strikethrough styled ... review all existing open or resolved workable items if they are obsolete - not valid any more ... There MUST NOT be any mistake! No bluff is allowed of any kind!"*

The §11.4.15 Status closed-set is extended with a terminal Obsolete (→ Fixed.md) value (orthogonal to Type per §11.4.16). Obsolescence reasons (closed vocabulary): superseded-by-design-change | superseded-by-later-mandate | feature-removed | duplicate-of | unsupported-topology. Every Obsolete heading MUST carry an ****obsolete-Details:**** line (Since + Reason + Superseding-item + Triple-check evidence) within 8 non-blank lines. The §11.4.23 colorizer adds a cell-status-obsolete class — light-gray #E0E0E0 background + strikethrough description. Audit cadence: every release-gate sweep per §11.4.40 + §11.4.42; triple-check is non-negotiable per the operator mandate. Composes with §11.4.15 / §11.4.16 / §11.4.19 / §11.4.21 / §11.4.23 / §11.4.33 / §11.4.34 / §11.4.40 / §11.4.42 / §11.4.66 / §11.4.71. Gates CM-COVENANT-114-90-PROPAGATION + CM-ITEM-OBSOLETE-DETAILS + CM-OBSOLETE-COLORIZER-WIRED + paired §1.1 mutations.

Cascade requirement: This anchor (verbatim or by §11.4.90 reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Propagation gate CM-COVENANT-114-90-PROPAGATION; paired mutation strips the literal → gate FAILs. Release blocker. **Canonical authority:** constitution submodule Constitution.md §11.4.90 for the full mandate.

§11.4.91 — Summary-Doc Clarity Mandate (cascaded from constitution submodule §11.4.91)

Verbatim user mandate (2026-05-27): *"Summary docs - Issues_Summary some not clear one line descriptions - like 'Composes with' ... For each workable item we MUST HAVE clearly understandable meaning ... every team member can clearly understand what that particular workable item is exactly about! There cannot be misunderstanding or unclarity of any kind and no bluff allowed!"*

Every summary entry (Issues_Summary, Fixed_Summary, README doc-link, Status_Summary pages 1+2, all one-liners) MUST contain a self-contained meaningful description ≥ 6 words OR ≥ 40 chars naming SUBJECT + PROBLEM/GOAL. Forbidden one-liner anti-patterns: section labels (Composes with, Closure criteria, Fix direction, etc.); bare metadata fragments (Critical, Bug, In progress, etc.); section-marker echoes; a \$-letter alone. Generators (generate_issues_summary.sh / generate_fixed_summary.sh / update_readme_doc_links.sh / generate_status_summary.sh) MUST extract from the H1/H2 heading line per the §11.4.54 ATM-NNN convention, NEVER from arbitrary downstream text, and MUST refuse anti-pattern rows — emitting a (MISSING DESCRIPTION – fix source heading) placeholder with visual highlight. Gate CM-SUMMARY-CLARITY-DESCRIPTIONS scans every summary; an anti-pattern match = FAIL. Audit cadence: every §11.4.40 + §11.4.42 sweep.

Cascade requirement: This anchor (verbatim or by §11.4.91 reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Propagation gate CM-COVENANT-114-91-PROPAGATION; paired mutation strips the literal → gate FAILs. Release blocker. **Canonical authority:** constitution submodule Constitution.md §11.4.91 for the full mandate.

§11.4.92 — Multi-Pass Change-Evaluation Discipline (cascaded from constitution submodule §11.4.92)

Verbatim user mandate (2026-05-27): *"Every change to the project or codebase we do MUST BE evaluated in several passes and in in-depth analysis for potential new issues or problems it can introduce! ... no bluff of any kind! After we do change or set of changes this mandatory steps MUST BE taken!"*

Every non-trivial change MUST pass a 5-pass evaluation BEFORE it is commit-ready: **(Pass 1)** main-task verification — change achieves the stated goal, captured-evidence per §11.4.5/§11.4.69; **(Pass 2)** regression-blast-radius analysis — enumerate every direct dependency, demonstrate no contract break; **(Pass 3)** cross-feature interaction analysis — audit parallel features sharing state/timing/hardware/shell environment; **(Pass 4)** deep-research validation per §11.4.8 — external precedent OR "NO external solution found — original work" + CodeGraph queries per §11.4.78/§11.4.79; **(Pass 5)** anti-bluff confirmation per §11.4 / §11.4.1 / §11.4.6 / §11.4.27 / §11.4.50 / §11.4.52 / §11.4.69 / §11.4.83 — no new bluff surface introduced. Each pass is documented (commit footers OR docs/ entries OR qa-results/ evidence). Only after all 5 passes complete may commit/push/test/release proceed. Trivial exemption: typo / revision-bump / MD-export-regen IF zero source touched AND the commit message cites the exemption explicitly. Gates CM-COVENANT-114-92-PROPAGATION + CM-MULTI-PASS-EVALUATION-EVIDENCE + paired §1.1 mutations.

Cascade requirement: This anchor (verbatim or by §11.4.92 reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Propagation gate CM-COVENANT-114-92-PROPAGATION; paired mutation strips the literal → gate FAILs. Release blocker. **Canonical authority:** constitution submodule Constitution.md §11.4.92 for the full mandate.

§11.4.93 — SQLite-Backed Single-Source-of-Truth for Workable Items (cascaded from constitution submodule §11.4.93)

Verbatim user mandate (2026-05-27): *"There MUST be single source of truth for all of our workable items - SQLite database ... proper scripts (we recommend Go programs) ... reduce a chance for sync to be broken ... generate always all docs from DB or to re-generate Db from all docs we have in opposite direction"*

The text-based Issues/Fixed/Summary/CONTINUATION constellation is converted to a SQLite-DB-backed single source of truth. Schema mandatory tables: items (atm_id PK + Type + Status incl. Obsolete + Severity + title + description ≥40 chars + created/modified + composes_with JSON + current_location); item_history (append-only audit per §11.4.34 By/Reason/Evidence); obsolete_details (§11.4.90); operator_block_details (§11.4.21); firebase_metadata (§11.4.47); meta (schema_version + last_sync + integrity_hash). A Go binary at cmd/workable-items/ provides sync md-to-db / db-to-md / diff / validate / add / close; bidirectional regen is byte-identical round-trip (closed-set whitespace/section-order

tolerance). `commit_all.sh` refuses on non-empty diff;
`sync_issues_docs.sh` invokes the Go binary; pre-build runs `workable-items validate`. Anti-bluff: unit + integration + stress (1000-row insert + 10 concurrent writers) + chaos (mid-write SIGKILL + corrupt-DB recovery + disk-full) + paired §1.1 mutation + HelixQA Challenge CME-WORKABLE-ITEMS-001. The Go binary lives in the constitution submodule (`constitution/scripts/workable-items/`) per §11.4.74. Gates CM-COVENANT-114-93-PROPAGATION + CM-WORKABLE-ITEMS-DB-PRESENT + CM-WORKABLE-ITEMS-MD-DB-IN-SYNC + paired §1.1 mutations. (NOTE: the DB tracking rule is AMENDED by §11.4.95 — DB is TRACKED, not gitignored.)

Cascade requirement: This anchor (verbatim or by §11.4.93 reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Propagation gate CM-COVENANT-114-93-PROPAGATION; paired mutation strips the literal → gate FAILs. Release blocker — text-based-only trackers are a §11.4 PASS-bluff at the data-architecture layer. **Canonical authority:** constitution submodule Constitution.md §11.4.93 for the full mandate.

§11.4.94 — Zero-Idle Priority-First Parallel-By-Default Operating Mode (cascaded from constitution submodule §11.4.94)

Verbatim user mandate (2026-05-27): "We MUST NEVER sit iddle / wait or sleep if there is possibility for us to work on something ... Always check if there is a possibility to work on something while we are not working actively on something! Pick always by priority - most critical workable items and other tasks MUST BE done first! ... Stay still / iddle if nothing is left to be done at all or waiting for something that is blocking us / you!!!"

§11.4.94 binds §11.4.20 + §11.4.42 + §11.4.58 + §11.4.70 + §11.4.72 + §11.4.82 + §11.4.87 + §11.4.88 + §11.4.89 into a single always-on enforcement: (A) idle ONLY when every queued item is genuinely blocked on an external dependency (hardware / network upstream / build/test completion the conductor cannot accelerate) OR operator STOP OR §12 host-safety — "don't see what to do" is NEVER valid; (B) before ANY wake/sleep the conductor MUST survey parallel-work feasibility per §11.4.42 + §11.4.72 + §11.4.87, identify non-contending items, and dispatch in parallel per §11.4.20/§11.4.70 (subagent) + §11.4.58 (Pwu disjoint scope) + §11.4.89 (background long tests); (C) priority order MANDATORY — pick highest-severity + §11.4.72 audio-first the conductor can autonomously progress; (D) subagent-driven default for non-trivial; (E) background default for >30 s wall-clock

work via nohup+disown; (F) stability-preserving (composes with §11.4.92 multi-pass + §11.4.84 quiescence + §12.6–§12.9 host safety); (G) progress updates surfaced at milestone boundaries. Gates CM-COVENANT-114-94-PROPAGATION + CM-PARALLEL-WORK-AUDIT + paired §1.1 mutations.

Cascade requirement: This anchor (verbatim or by §11.4.94 reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Propagation gate CM-COVENANT-114-94-PROPAGATION; paired mutation strips the literal → gate FAILs. Release blocker. **Canonical authority:** constitution submodule Constitution.md §11.4.94 for the full mandate.

§11.4.96 — Safe-Parallel-Work-With-Long-Build Catalogue + Mandate (cascaded from constitution submodule §11.4.96)

Verbatim user mandate (2026-05-27): *"Are there except AOSP build process any other active jobs being done at the moment? Can we work on something in parallel while build is in progress so we slowly cleanup our slate? ... do as much as possible work in background in parallel with main work stream and oreferably using subagents-driven approach!"*

An operational catalogue for the canonical long-running workload (multi-hour containerised build per §12.9). **SAFE during build:** (A) MD/docs work; (B) generator/helper script work under scripts/; (C) pre-build + meta-test gate authoring + paired §1.1 mutations; (D) on-device test scripts; (E) constitution submodule edits + push; (F) any submodule commit + push per §11.4.88; (G) read-only live-ADB probes (dumpsys/getprop/cat /proc/.../screencap/logcat); (H) subagent dispatch per §11.4.20/§11.4.70 + §11.4.84 quiescence; (I) web research + external API queries with §11.4.10 credentials; (J) workable-items DB ops per §11.4.93+§11.4.95; (K) backgrounded pre-build + meta-test execution per §11.4.89. **UNSAFE during build:** (α) git checkout/reset --hard/clean -df on the source tree (use git worktree); (β) mass file deletes/renames under built source trees; (γ) submodule pointer updates affecting built artefacts; (δ) out/ mutations; (ε) make clean/m clobber/rm -rf out/; (ζ) container destruction; (η) disk-filling breaching §12.9 free-space minimum; (θ) §12 host-session-safety breaches. Conductor responsibility: before EVERY pause point during a long build, consult the catalogue, identify (A)-(K) queue items per §11.4.42+§11.4.72, and dispatch ≥1 per §11.4.20/§11.4.70 sub-agent default + §11.4.89 background. "Build running, nothing else

to do" is NEVER true per §11.4.94+§11.4.96. Gates CM-COVENANT-114-96-PROPAGATION + CM-PARALLEL-WORK-DURING-BUILD-AUDIT + paired §1.1 mutations.

Cascade requirement: This anchor (verbatim or by §11.4.96 reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Propagation gate CM-COVENANT-114-96-PROPAGATION; paired mutation strips the literal → gate FAILs. Release blocker. **Canonical authority:** constitution submodule Constitution.md §11.4.96 for the full mandate.

§11.4.97 — Maximum-Use-of-Idle-Time + Progress-Update Cadence (cascaded from constitution submodule §11.4.97)

Verbatim user mandate (2026-05-27): "keep it working, we should do as much as possible, if not it all but as much as we can as long as there is iddle time! it MUST be used! ... keep us updated about all progress and all phisycal proofs and gathered data as you progress through all open workable items!"

Operating-mode capstone strengthening §11.4.87 + §11.4.94 + §11.4.96: (A) every minute of conductor idle time during which work could autonomously progress AND is not genuinely blocked = a §11.4.97 violation; "as much as possible, if not it all but as much as we can" is operative — dispatch CONTINUOUSLY through the entire idle window, not just at scheduled wakes; (B) progress-update cadence — emit an operator-facing 1-line update at every commit landed / subagent return / constitutional anchor / captured evidence / milestone closure, no operator prompt required; (C) continuous physical-proof gathering per §11.4.5 + §11.4.6 + §11.4.69 — every autonomous closure cites captured-evidence (evidence path goes into the §11.4.93 item_history.evidence_path when the DB lands); (D) composes with §11.4.5/6/13/20/27/42/50/52/69/70/72/83/85/87/88/89/94/96; (E) the idle-only-when-blocked closed-set is unchanged from §11.4.94(A). Gates CM-COVENANT-114-97-PROPAGATION + CM-IDLE-TIME-AUDIT + paired §1.1 mutations.

Cascade requirement: This anchor (verbatim or by §11.4.97 reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Propagation gate CM-COVENANT-114-97-PROPAGATION; paired mutation strips the literal → gate FAILs. Release blocker. **Canonical authority:** constitution submodule Constitution.md §11.4.97 for the full mandate.

§11.4.95 — Workable-Items SQLite DB Is TRACKED in Git, NEVER Gitignored (cascaded from constitution submodule §11.4.95)

Verbatim user mandate (2026-05-27): *"We shall not Git ignore our workable items SQLite DB since it is our single source of truth ... workable items SQLite DB regularly committed and pushed to all upstreams!"*

§11.4.93's earlier "gitignored per §11.4.30" clause is AMENDED — the DB at docs/workable_items.db is TRACKED in git, NEVER gitignored. It IS authoritative source data, NOT a build artefact. Every workable-items sync md-to-db that mutates state MUST stage + commit + push the DB alongside the MD regen per §11.4.19 atomic-move + §2.1 multi-upstream push. A WAL-checkpoint (PRAGMA wal_checkpoint(TRUNCATE)) is required before commit-stage so the transient .db-wal + .db-shm sidecars (gitignored per §11.4.30) are safely discardable. The §11.4.77 regeneration mechanism does NOT apply — the DB IS the source. Destructive DB ops require §9.2 hardlinked-backup + operator authorization; §11.4.41 force-push merge-first applies if DB history ever needs rewrite. Gates CM-COVENANT-114-95-PROPAGATION + CM-WORKABLE-ITEMS-DB-TRACKED + paired §1.1 mutation.

Cascade requirement: This anchor (verbatim or by §11.4.95 reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Propagation gate CM-COVENANT-114-95-PROPAGATION; paired mutation strips the literal → gate FAILs. Release blocker. **Canonical authority:** constitution submodule Constitution.md §11.4.95 for the full mandate.